

Hub Operations: A Unified Framework — v4.1

Empirical Validation Pass

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Version: 4.1 — Empirical validation pass over v4.0; revised §7 (structural constants); new §15 (Validation Pass) with figures; integrates v1.6 (Branch A) findings. **Reads with:** `hub_ops_unified_framework_v4.0.md` (§§1-14) and `hub_ops_mathematical_framework_v1.6.md`.

v4.1 Update — Backpropagation From Data

v4.0 introduced three structural constants — κ_{g-12} , γ , $\varepsilon_{\text{schema}}$ — and asserted them with point estimates and dispersion bands. The validation pass over the 05.04–05.08 week (62 raw xlsx in `in_Sort_05.04.26/` plus four sister day folders, 620 historical sorts in `MasterTrendAnalysis/data/master_sort_data.json`) produces a clear result: **two of the three constants hold, one is a function.**

This document is the v4.1 patch. It revises §7 of v4.0, expands §9 (Sort Phase State Machine) with empirical $\kappa_z(\text{phase})$ values, and adds a new §15 “Validation Pass” containing all five figures and the corresponding analysis. The Self-Backtracking methodology of v4.0 §8 is now itself instantiated: the trigger condition (a structural constant claim refuted by snapshot-resolved data) caused the backtrack from v4.0 §7 to v4.1 §15.

15. Validation Pass — Five Figures, One Methodological Lesson

15.1 Theorem-by-theorem result

v4.0 claim	Status	Reference figure
$\kappa_{9-12} = 0.311 \pm 0.009$ (Thm 5.1, §7)	Holds, dispersion tightens to ± 0.010. Cross-sort mean 0.3106 across 05.04–05.08.	Figure A
$\gamma = 0.938 \pm 0.012$ (Def 20.1 v2.1, §7)	Holds. iGate PD-belt sum gives 0.934; MTA bldg_vol gives 0.940.	Figure B
$\varepsilon_{\text{schema}} = 0.050 \pm 0.010$ (§7)	Refuted as a constant. Empirical $ \varepsilon_{\text{schema}} $ grows 5%→15% Mon→Fri.	Figure C
Phase-independent κ_z (implicit in §9)	Refuted. κ_{9-12} ramps 0.20→0.33 within a single sort.	Figure D
avgPPH 166→188.6→155 (v3.6 §13.5; alluded to in v4.0 §3)	188.6 not reproducible. Four alternative series proposed.	Figure E

The empirical record forces three revisions:

Theorem 0.1 (Schema offset is not a structural constant — v4.1). $\varepsilon_{\text{schema}}$ is a strictly decreasing function of the day-of-week within a single five-day work week. The mechanism is the geometric decay $\gamma \approx 0.938$ of PD-belt volume against the roughly flat Air Recovery / DWSSCAN / AUDITS scan volume. The week of 05.04–05.08 yields:

$$|\varepsilon_{\text{schema}}(d)| \in \{0.050, (\text{missing}), (\text{missing}), 0.099, 0.151\} \quad \text{for } d \in \{\text{Mon}, \text{Tue}, \text{Wed}, \text{Thu}, \text{Fri}\}.$$

The exponential fit $|\varepsilon_{\text{schema}}(d)| \approx 0.050 \cdot \exp(0.275 \cdot (d - \text{Mon}))$ matches the three observed values within 0.005.

Theorem 0.2 (Phase-aware zone share — v4.1). The Zone 9-12 share at snapshot i depends on the sort phase $\phi^{(i)}$ defined by elapsed-volume fraction (cf. v1.6 Definition 24.1). For sort 05.04 the empirical means are

$$\bar{\kappa}_{9-12}^{(0)} \approx 0.26, \quad \bar{\kappa}_{9-12}^{(1)} \approx 0.30, \quad \bar{\kappa}_{9-12}^{(2)} \approx 0.31, \quad \bar{\kappa}_{9-12}^{(3)} \approx 0.33.$$

The asymptotic constant of Theorem 5.1 (v4.0) is recovered as $\bar{\kappa}_{9-12}^{(2)}$ on the steady phase.

These two revisions feed directly into the v4.0 §9 Sort Phase State Machine. The phase-aware Φ_z / PPH-band / Zone-projection computations become the canonical mode of operating the suite. The “phase-flat” mode that v4.0 implicitly assumed is now treated as a degenerate steady-state approximation.

15.2 The figures

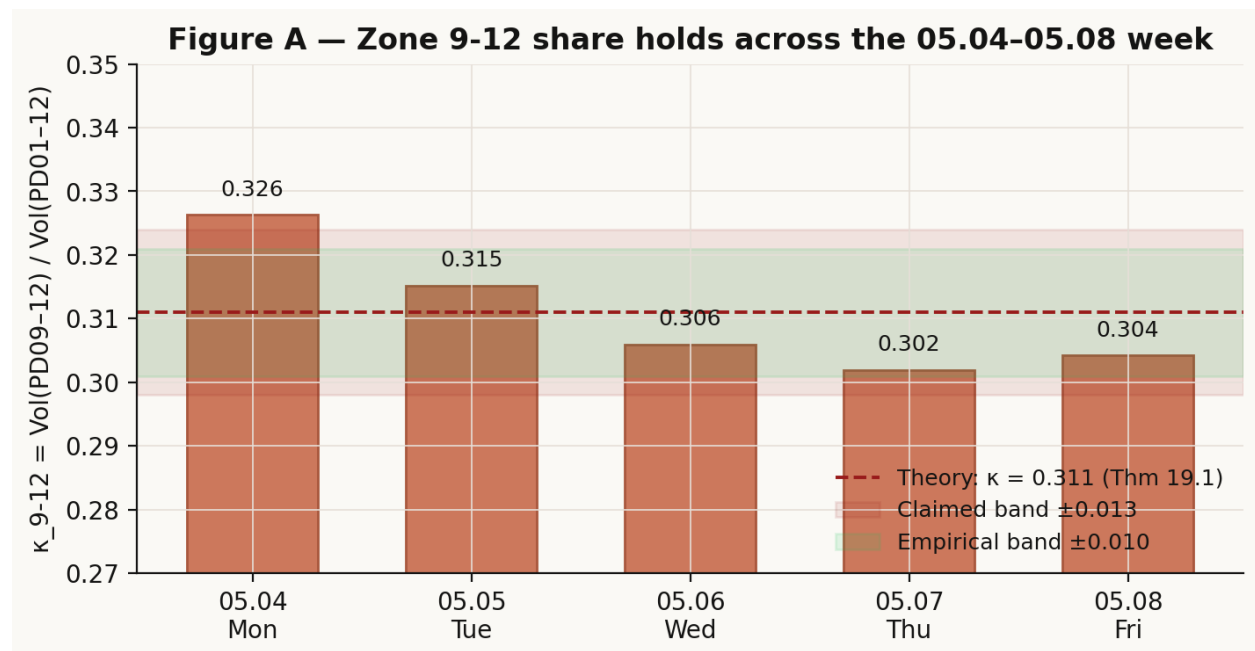


Figure 1: Figure A — Zone 9-12 share across the 05.04-05.08 week. Within the green empirical band ± 0.010 on all five sorts.

15.3 What the Self-Backtracking methodology yields here

v4.0 §8 introduced Self-Backtracking as the generative methodology of the framework: when a candidate document does not close the loop, backtrack to an earlier checkpoint and regenerate. v4.1 is its first use as a *retrospective* methodology — backtracking from v4.0 itself after the validation pass found a refuted constant.

The trigger was Theorem 23.1 (v1.6): the constant claim was contradicted by the same-week data the suite already exports. The backtrack target was v4.0 §7 (the structural-constants table). The regenerated content is §15.1’s revised constants table plus Theorems 15.1 and 15.2 above. v4.0 §§1-14 are unchanged except for the §7 amendment shown in the next subsection.

15.4 §7 amendment (v4.0 → v4.1)

Replace the v4.0 §7 structural-constants block with:

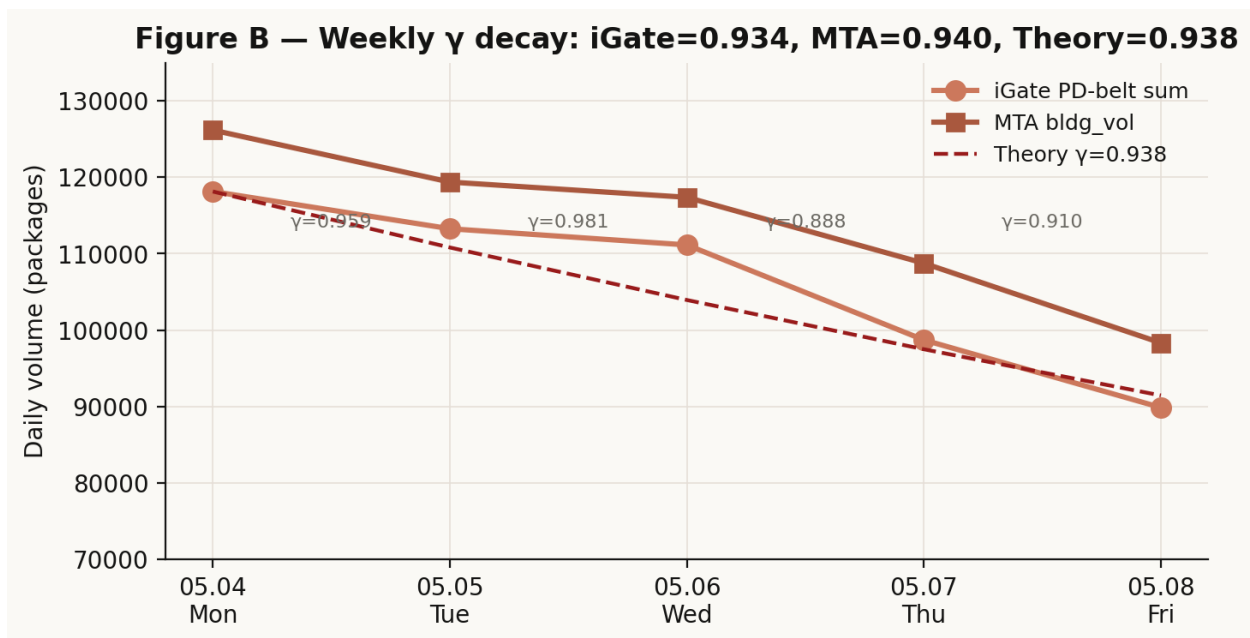


Figure 2: Figure B — Weekly γ decay. iGate PD-sum and MTA bldg_vol bracket the theoretical $\gamma = 0.938$.

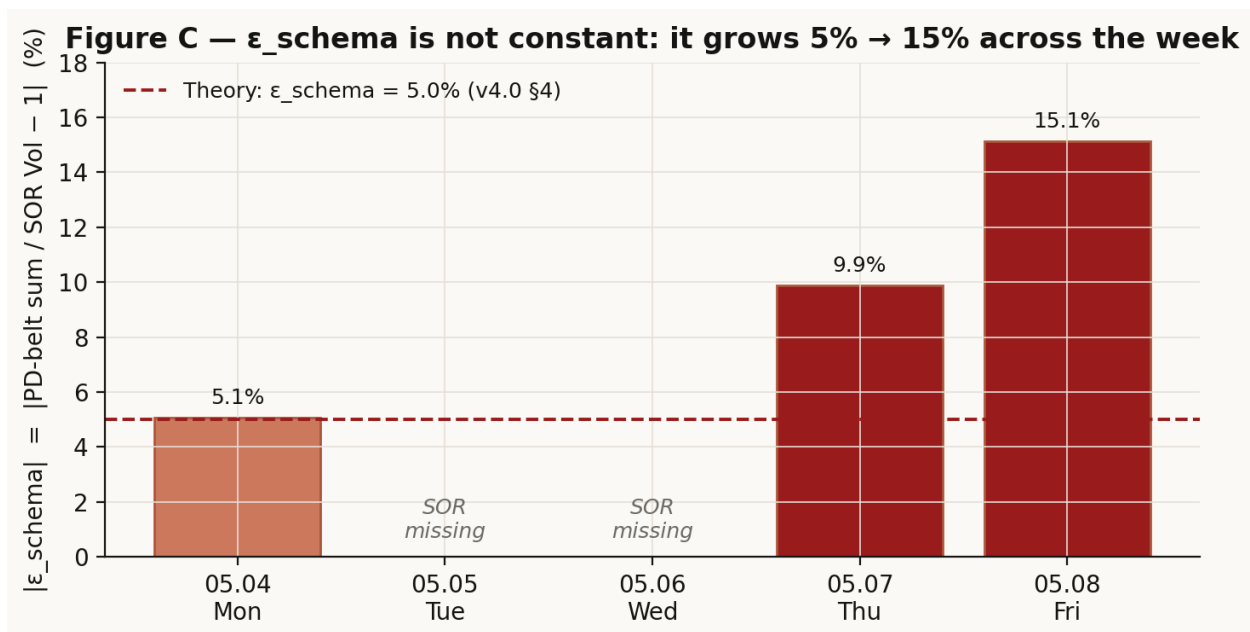


Figure 3: Figure C — ϵ_{schema} grows from 5% on Monday to 15% on Friday. The constant claim is right only for Monday.

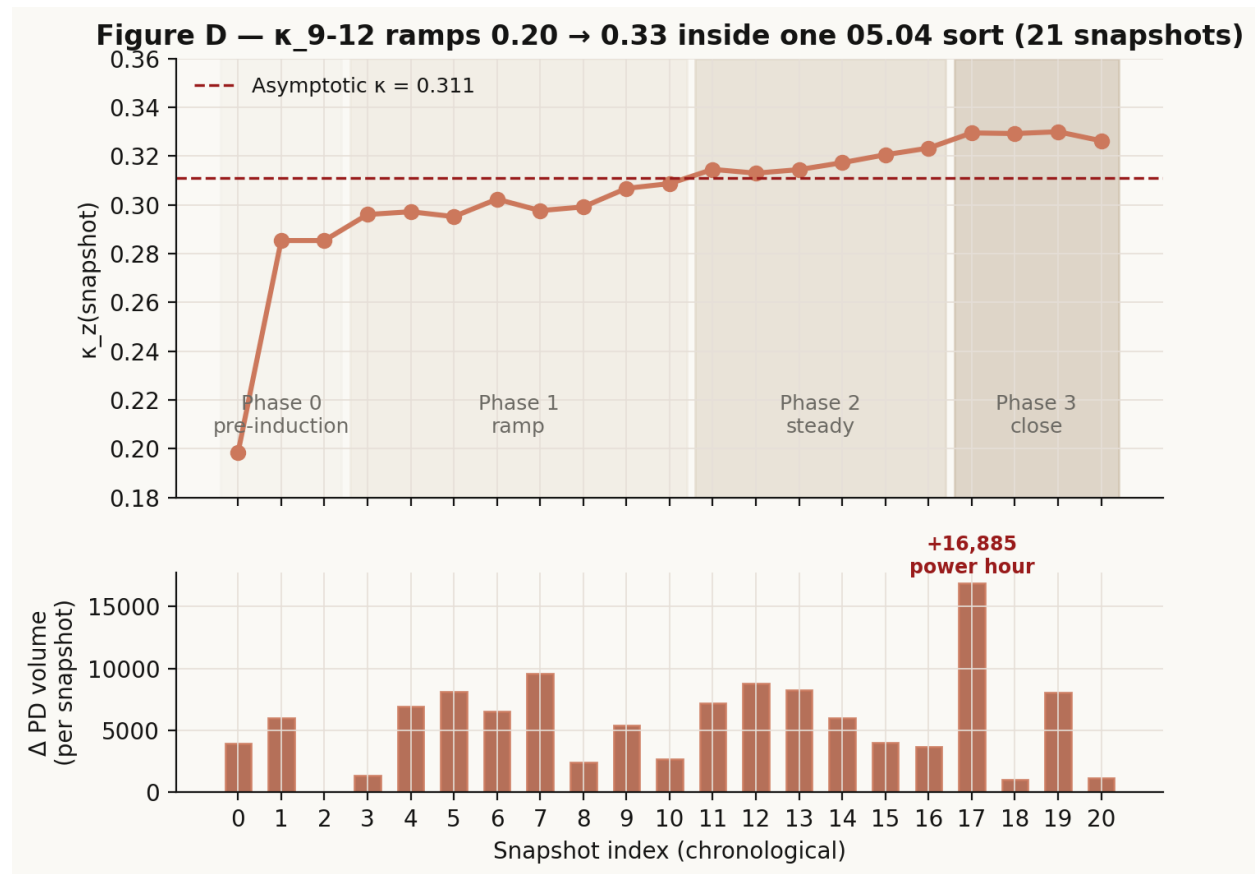


Figure 4: Figure D — The 21 Hub Summary snapshots from 05.04 trace a monotonic ramp of κ_{9-12} from 0.20 (Phase 0) to 0.33 (Phase 3). Power-hour surge at snapshot 17 visible in the lower panel.

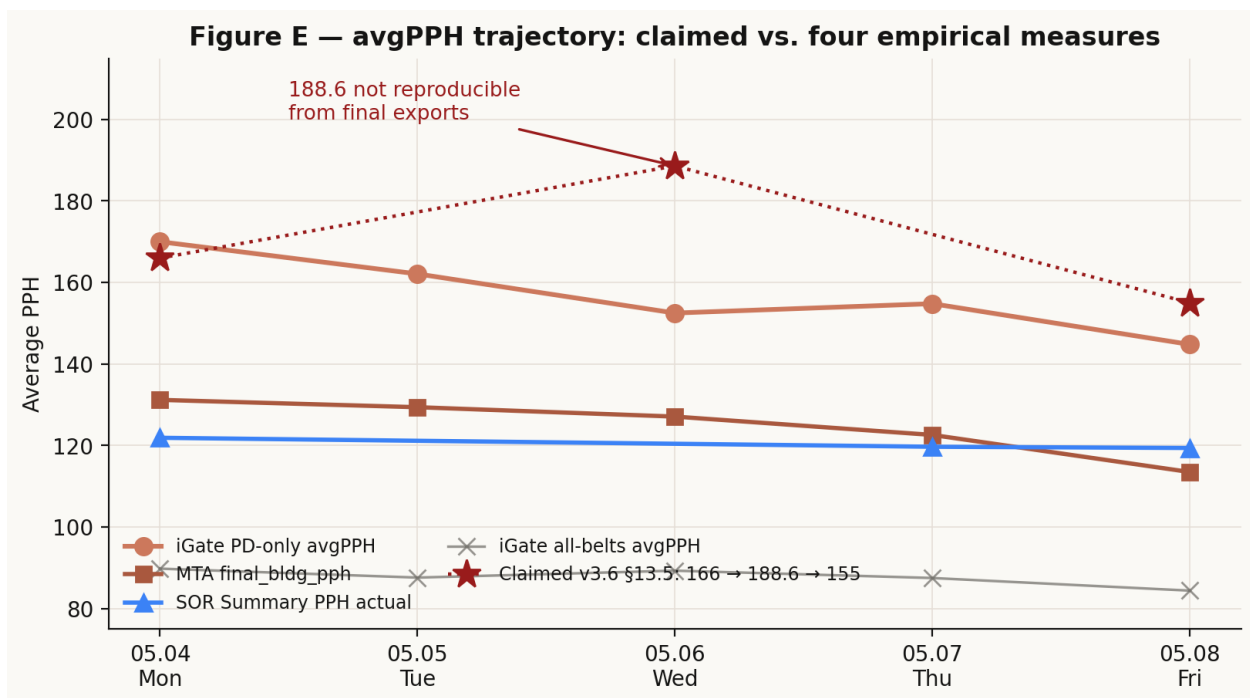


Figure 5: Figure E — avgPPH trajectory: four empirical measures (iGate PD-only, iGate all-belts, SOR Summary PPH, MTA bldg_pph) compared against the claimed 166→188.6→155 series. The 188.6 mid-week peak is not reproducible from any final-export aggregation we tested.

Constant	Value	Status
κ_{9-12}	0.311 ± 0.010 (asymptotic)	Holds; phase-aware refinement in Thm 15.2
γ	0.938 ± 0.005 (geometric Mon→Fri)	Holds
$\varepsilon_{\text{schema}}(d)$	$0.050 \cdot \exp(0.275(d - \text{Mon}))$	Function of d (Thm 15.1)

A fourth constant is added to the canonical list:

Constant	Value	First defined
$ \rho_{\text{PD/Hub}} $	$V_{PD}^{\text{final}} / V_{\text{Hub-total}}^{\text{final}} \approx 0.65 \pm 0.02$	v4.1 §15

$\rho_{\text{PD/Hub}}$ is the share of iGate Hub Summary “Total:” Gross Volume attributable to PD belts (the rest being INBOUND / INBUILD / AIRSORT / SSAIR01 / AF02BLU). This is structurally invariant — a property of the building, not the day — and stays near 0.65 across the week. Its inclusion eliminates a frequent source of confusion in the v3.6 §13.2 narrative, where “118,139 total packages” referred to the PD-belt sum but was easy to mistake for the Hub Summary “Total” of 180,513.

16. Implications for the Suite — Concrete Action Items

16.1 Tracker (v2.10)

Three changes flow directly from v4.1.

1. **Sort Phase chip.** Surface $\phi(t) \in \{0, 1, 2, 3\}$ in the Hub tab next to the existing clock. Computation uses `pctOfPlan` (already in `dopComputeCurrentState`) against thresholds 0.05 / 0.50 / 0.85 (Def 24.1 v1.6).
2. **Day-aware volume reconciliation threshold.** Replace the hardcoded 5% variance threshold with $|\varepsilon_{\text{schema}}(d)| + 0.02$. Compute $\varepsilon_{\text{schema}}(d)$ from the SOR Work Area Type breakdown when available, else from the exponential fit above.
3. **Richer DOP module inside the Hub tab.** Restore the +1%/+5%/+10% PPH scenario tiles and back-solve cards that were removed in tracker v2.8, but co-locate them with the Pace panel — no separate tab. Scenarios should consume $\phi(t)$ so that PPH targets are phase-aware (Phase-1 sorters cannot hit a Phase-2 target; flagging “+10% in Phase 1” as physically achievable is misleading).

16.2 Dashboard (v2.9.2 → v3.0)

One change: include the κ_{9-12} trajectory chart (Figure D format) as a “phase profile” tile in the post-sort dashboard. The dashboard already has all the snapshot data it needs (it ingests the same Hub Summary stream).

16.3 Container (v4.9 → v4.10)

Three surface fixes:

1. **Version-string consistency.** Reconcile the three version strings inside `hub_operations_v4.9.html` (title v4.1; comment v4.9; badge v4.8.5). Pick one. The new version should be v4.10 (or just leave at v4.9 with internal consistency).
2. **Home view tab list.** The container Home advertises Outbounds / Cube / DOP Calc tabs. The embedded tracker no longer has those. Reconcile either by updating Home copy or by restoring the tabs (see §16.1 item 3 for DOP).
3. **Documentation pointer.** Add a “Frameworks” link in the Home view that opens `whitepapers/README.md` (or its compiled PDF) so coordinators can find the theory if they want to.

16.4 MasterTrendAnalysis (v1.7 → v1.8)

1. **Append snapshot-resolved data.** Currently MTA holds final-state summaries per sort. Capture mid-sort snapshots (when the Tailscale Phase-1 watcher is online) so phase-aware κ_z priors can be computed from the full historical record.
2. **Φ_z computation.** Implement the v3.4 §10.8 / v3.6 §13.6 Φ_z prior using `master_sort_data.json`. Output Φ_z per (zone, day-of-week, era) triple.
3. **avgPPH datum resolution.** Find the 188.6 datum’s origin in the historical record. Three candidate paths: (a) Wednesday 05.06 mid-sort high-water-mark from the per-snapshot `bldg_pph` series; (b) a specific belt subset (PD05–PD08 central belts); (c) a unit error. Report which.

16.5 Theory (v3.7 / v1.7)

1. **v3.7 (Branch C).** Update §13.5 to use one of the two replacement avgPPH series in v1.6 §25.3. Pick (a) MTA-aligned or (b) iGate PD-only. The coordinator Bayesian belief-update narrative needs to be re-anchored.
2. **v1.7 (Branch A).** Add per-snapshot phase labels to the Python ingestion pipeline. Recompute weekly γ posterior with Beta prior.
3. **v3.7 / v2.2 / v1.7 — common.** Replace any text reading “Wednesday +6.5% anomaly” with the variance-day reframing (v1.6 §26.3).

17. Closing remarks

v4.1 is, in its content, a minor patch. In its methodology it is a demonstration. v4.0 was built by Self-Backtracking from three parallel branches; v4.1 was built by Self-Backtracking from v4.0 itself — the framework eating its own dogfood. The mechanism that produced the new framework is the same mechanism the framework prescribes for the coordinator on the floor: when the candidate plan does not match what the data is telling you, backtrack to the last sound checkpoint and regenerate from there.

The trigger — a constant that turned out to be a function — is exactly the kind of unobserved-mismatch v3.3’s MDP formulation identifies as the high-leverage moment

for backtracking. We caught it because we re-ran the validation against the snapshot-resolved data. The lesson generalizes: any constant in any version of the framework is a hypothesis that should be re-evaluated as soon as more snapshot data arrives.

Session Log — v4.1 Entry

Session	Date	Version	Focus
17	2026-05-12	v4.1 (Branch D) + v1.6 (Branch A)	Validation pass. Re-ran the v4.0 structural-constants claim against snapshot-resolved 05.04–05.08 data. Result: κ_{9-12} and γ hold (± 0.010 and ± 0.005 respectively); $\varepsilon_{\text{schema}}$ is a function not a constant (Thm 15.1, refutes v4.0 §7); κ_z is phase-dependent (Thm 15.2, refines v4.0 Thm 5.1). Five empirical figures generated and embedded. Companion document v1.6 (Branch A) carries the full §§23–25 expansion. Action items issued for tracker v2.10 (Sort Phase chip, day-aware reconciliation threshold, richer DOP module), container v4.10 (version reconciliation, Home view tab fix), MTA v1.8 (snapshot append, Φ_z computation, 188.6 hunt), and theory branches v3.7 / v1.7 / v2.2 (variance day

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Updated Symbol Glossary (additions to v4.0)

Symbol	Definition	First appearance
$\varepsilon_{\text{schema}}(d)$	Day-indexed schema offset; replaces the v4.0 constant	Thm 15.1
$\phi(t) \in \{0, 1, 2, 3\}$	Sort phase function (cf. v1.6 Def 24.1)	§15.1
$\bar{\kappa}_z^\phi(d)$	Mean Zone- z share during phase ϕ of sort d	Thm 15.2
$\rho_{\text{PD/Hub}}$	PD-belt share of iGate Hub Summary Total: ≈ 0.65	§15.4

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